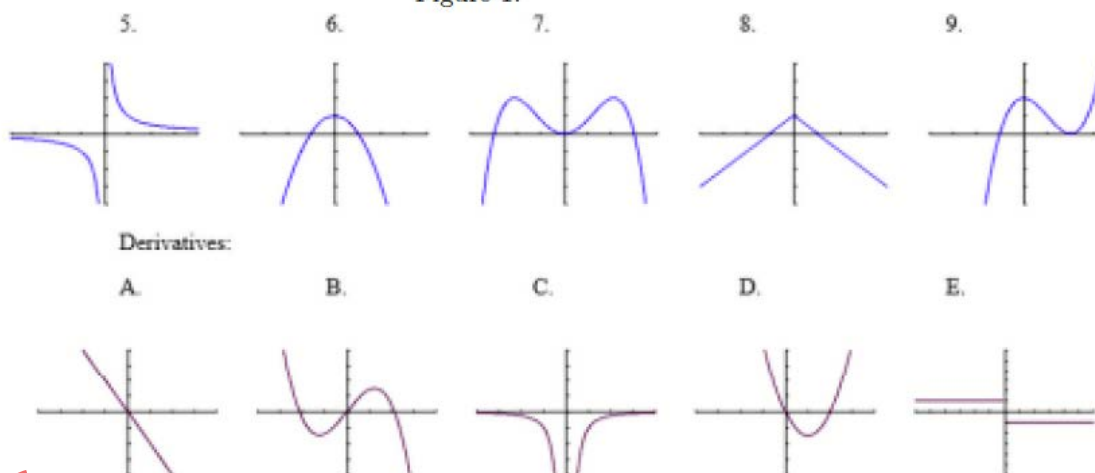


1. The position function of a moving particle is given by $s(t) = 2t^2 + 2$, where s is in meters and t is in seconds.

- Find the average velocity of the particle over the time interval $[1, 2]$.
- Find the particle's instantaneous velocity at time $t = 1$ s.
- Draw the position function $s(t)$, and velocity function $v(t)$ on the same screen.

2. Match each of the following graphs(5 – 9) with its derivative graph(A – E). [3]

Figure 1:



3. Consider the functions $f(x) = x^2$ and $g(x) = x + 1$.

- Find $y = f(g(x))$ and y' using the chain rule.
- Find the equation of the tangent line to the graph of the function $y = f(g(x))$ at $x = 0$.
- Draw the graph of the function $y = f(g(x))$ with the tangent line at $x = 0$.

4.

Consider the function over the given interval.

$$f(x) = 2x - 6.$$

- Use an area formula from geometry to find the area function $A(x)$ that gives the area between the graph of the function $f(x)$ and the interval $[4, x]$.
- Confirm that $A'(x) = f(x)$.
- Also, verify your answer by direct integration.

5.

- Given that $\frac{dy}{dx} = \frac{5}{\sqrt{2x+1}}$ and $y = 7$ when $x = 4$. Express y in terms of x .

- The gradient of a function $y = f(x)$ is given by $x\sqrt{x} + 2$. If the curve passes through the point $(4, 1)$, find the equation of the function.

6. (a) Draw the graphs of $y = x - 5$, $y = -x + 5$, $x = 0$ and $x = 10$.
 (b) Find the area of the region enclosed by the graphs in part (a).
 (c) Draw the graphs of $x = y^2$ and $y = -x + 6$.
 (d) Find the area of the region enclosed by the graphs in part (c).

7. Evaluate the following integrals (answer any 4):

- (a) $\int 3t^5 \sqrt{5+t^6} dt$ (e) $\int_0^1 \frac{8x}{1+16x^2} dx$
 (b) $\int \frac{\csc^2 \sqrt{x}}{\sqrt{x}} dx$ (f) $\int \frac{x^5+x^2+x}{x} dx$
 (c) $\int x^2 \cos x dx$ (g) $\int x^7 e^{-4x^8} dx$
 (d) $\int \cos^4 x \sin x dx$

8. Consider the function

$$f(x) = \begin{cases} |x-6| & \text{if } x \geq 0 \\ 6 & \text{if } -8 \leq x < 0. \end{cases}$$

Use $f(x)$ to evaluate the following integrals:

- (a) $\int_{-8}^0 f(x) dx$ (b) $\int_0^{10} f(x) dx$

9.

Sketch the region whose signed area is represented by the definite integral and evaluate the integral Using an appropriate formula from geometry

- (a) $\int_2^7 (2x-8) dx$
 (b) $\int_{-7}^0 \sqrt{49-x^2} dx$
 (c) $\int_{-3}^0 (2x-5) dx$

10. (a) The position of a particle moving along a straight line is given by $s(t) = 3t^3 - 5t^2 + 2t + 4$ (in meters). Find the expressions for velocity and acceleration as functions of time t .

- (b) The position of a particle is given by $s(t) = t^3 - 6t^2 + 9t + 2$. Determine the time when velocity is zero.
 (c) The displacement of a body is given by $s(t) = \sin(2t) + \cos(2t)$. Find its velocity and acceleration at $t = \pi/4$.
 (d) The acceleration of a moving body is given by $a(t) = 4t - 2$. If the initial velocity is 3 m/s, find the velocity and displacement functions.
 (e)

The equation of motion of a particle is $s = t^4 - 2t^3 + t^2 - t$, where s is in meters and t is in seconds.

- i. Find the velocity and acceleration as functions of t .
 ii. Find the acceleration after 1 s.
 11. Differentiate the following w.r.t. x

- i. $3e^{-x^2+1}$
 ii. $x \sin 2x + \sqrt{x}$
 iii. $\ln \sqrt{\frac{3x}{x^2-1}}$

(iv) $\frac{12}{\sqrt{(3x-x^2)^3}}$

(v) $\sqrt{x}(2x-3x^2)+1$

(vi) $(x^2+1)^2 e^{-2x}$

(viii) $\ln \sqrt[3]{3x^2-2x+1}$

12. Given that $y = f(x) = x^3 - 2x^2 + 3$.

(i) Find the value of x for which $f'(x) = 0$

(ii) Find the value of $f''(-1)$

13. Given that $y = \sin 2x + e^{-x} + x^2 + 1$, Evaluate $\frac{d^2y}{dx^2}$

14. Find the equations of the tangent and normal to the curve $y = -3x^2 + 6x + 5$ at $x = 1$

15. Find the interval for which the function $y = \frac{1}{3}x^3 - 3x^2 + 5x + 1$ is

(i) Increasing (ii) Decreasing

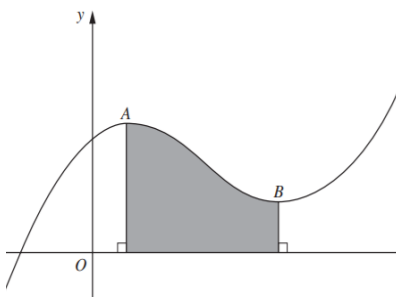
16. (i) If $x = \sin u$, and $u = \sqrt{t^2 + 1}$, find $\frac{dx}{dt}$.

(ii) Evaluate the integral $\int_{-2}^3 f(x) dx$, given that $f(x) = \begin{cases} x+5, & x < 0 \\ 5-x, & x \geq 0 \end{cases}$

17. (a) Find the area bounded by the curves $y = 2x$ and $y = x^2$

(b) Find the area bounded by the curves $y = x^3$, y axis and the lines $y=1$ and $y=4$.

18.



The diagram shows part of the curve $y = \frac{1}{3}x^3 - \frac{5}{2}x^2 + 4x + 1$. The curve has maximum point at A and a minimum point at B.

(i) Find the x -coordinate of point A and of point B

(ii) Find the area of the shaded region.

19.

(i) Evaluate $\int_1^2 x^2 \ln x$

(ii) By using appropriate substitution evaluate $\int_0^{\frac{\pi}{4}} \cos x \sqrt{\sin x + 3} dx$

20. Given that $y = x\sqrt{4x+12}$, show that $\frac{dy}{dx} = \frac{k(x+2)}{\sqrt{4x+12}}$, where k is a constant to be found.

Hence evaluate $\int_{-2}^6 \frac{3(x+2)}{\sqrt{4x+12}} dx$

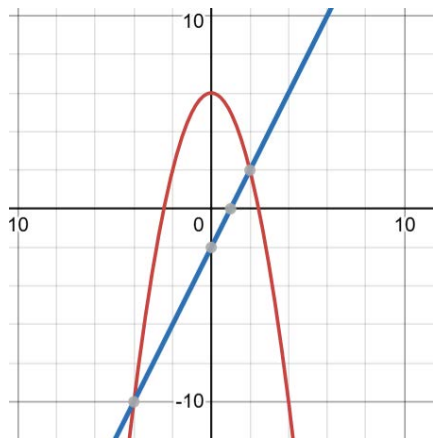
21. Given that $f(x) = \begin{cases} 2x+1 & 0 \leq x \leq 1 \\ 3x^2 & 1 < x \leq 2 \end{cases}$

(i) Sketch the graph of $f(x)$

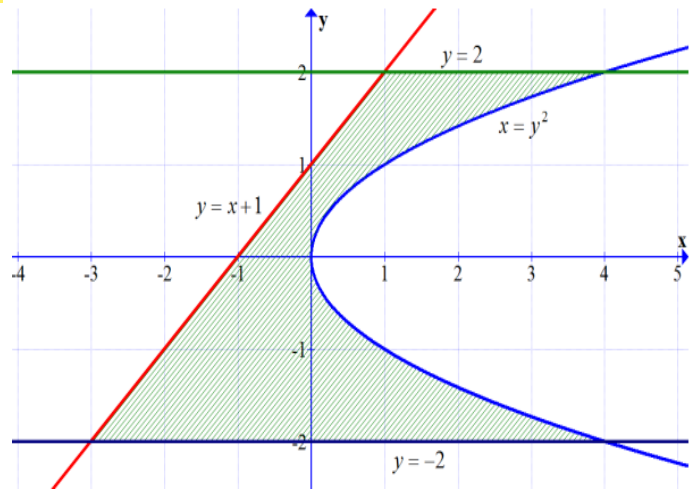
(ii) Evaluate $\int_0^2 f(x) dx$

22.

(a) Find the area bounded by the curve $y = 6 - x^2$ and the line $y = 2x - 2$



(b) Find the area of the shaded region



23. find the derivative curves of the following function

